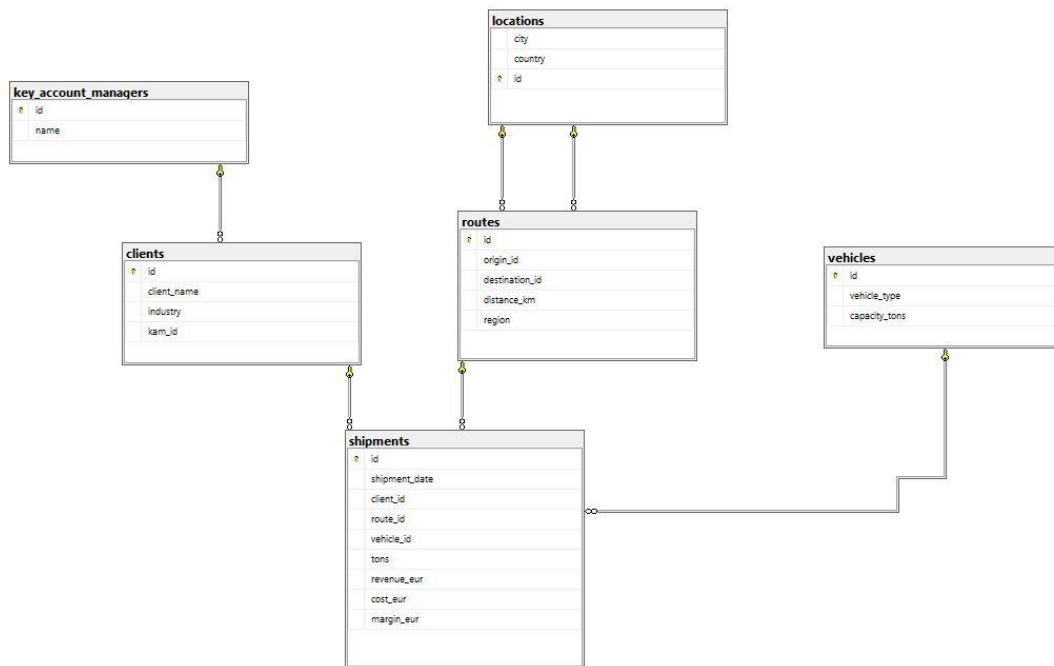


DIAGRAM



- 1. How many shipments have been completed in total?
- 2. What is the total revenue per client?
- 3. Which clients had more than 100 shipments in 2024?
- 4. What is the average margin (margin_eur) per route?
- 5. Which vehicle type is used most frequently?
- 6. What is the total margin per region within each industry?
- 7. Who are the top 5 clients by revenue per ton?
- 8. What is the average revenue per shipment by year and month?
- 9. Which Key Account Manager (KAM) manages the largest number of clients, and what is the margin per client for each KAM?
- 10. Which route (origin → destination) has the highest total distance (kilometers)?

- 11. What is the revenue ranking of clients within each industry?
- 12. What are the monthly revenue figures and the cumulative revenue (running total)?
- 13. What is the Year-over-Year (YoY) and Month-over-Month (MoM) revenue growth?
- 14. What is the profit margin percentage per client, the deviation from the average margin, and each client's revenue share in total revenue?
- 15. What is the most profitable route per region (Top 1 per region)?

/* =====

1. Koliko je ukupno pošiljki realizovano?

===== */

-- Izračunati ukupan broj zapisa u tabeli shipments.

SELECT

COUNT(id) AS Number_of_shipments

FROM

shipments;

	A	
1	Column1 ▾	
2	42000	
3		

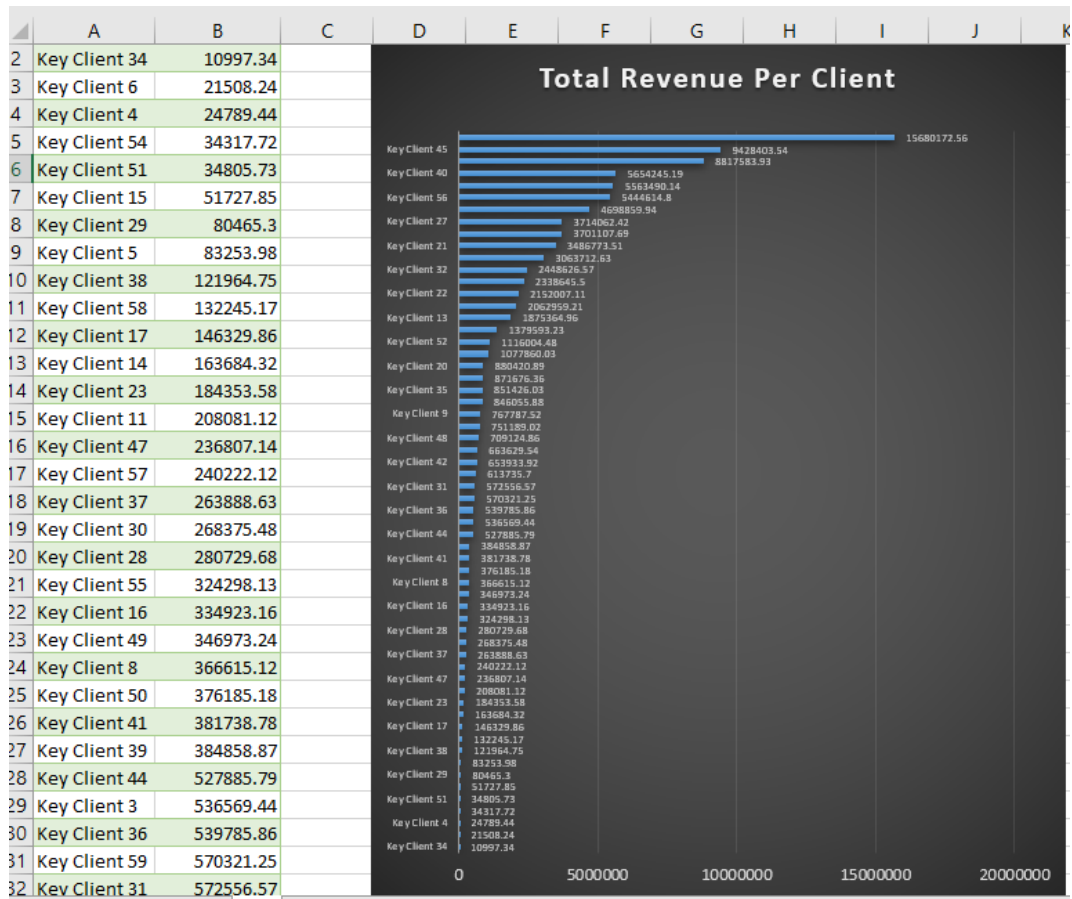
```
/* =====  
2. Koliki je ukupan prihod (revenue) po klijentu?  
===== */
```

```
-- Prikazati client_name i SUM(revenue_eur).
```

```
-- Potrebno je JOIN-ovati shipments i clients.
```

```
-- Grupisati po client_name.
```

```
SELECT  
    client_name,  
    ROUND(SUM(revenue_eur),2) AS Total_Revenue  
FROM  
    clients  
LEFT JOIN  
    shipments  
ON  
    clients.id = shipments.client_id  
GROUP BY  
    client_name  
ORDER BY  
    Total_Revenue DESC;
```



/* =====

3. Koji klijenti su imali više od 100 pošiljki u 2024. godini?

===== */

-- Izračunati COUNT po client_id.

-- Filtrirati koristeći HAVING > 100 i gde je YEAR = 2024.

-- Prikazati client_name i broj pošiljki.

SELECT

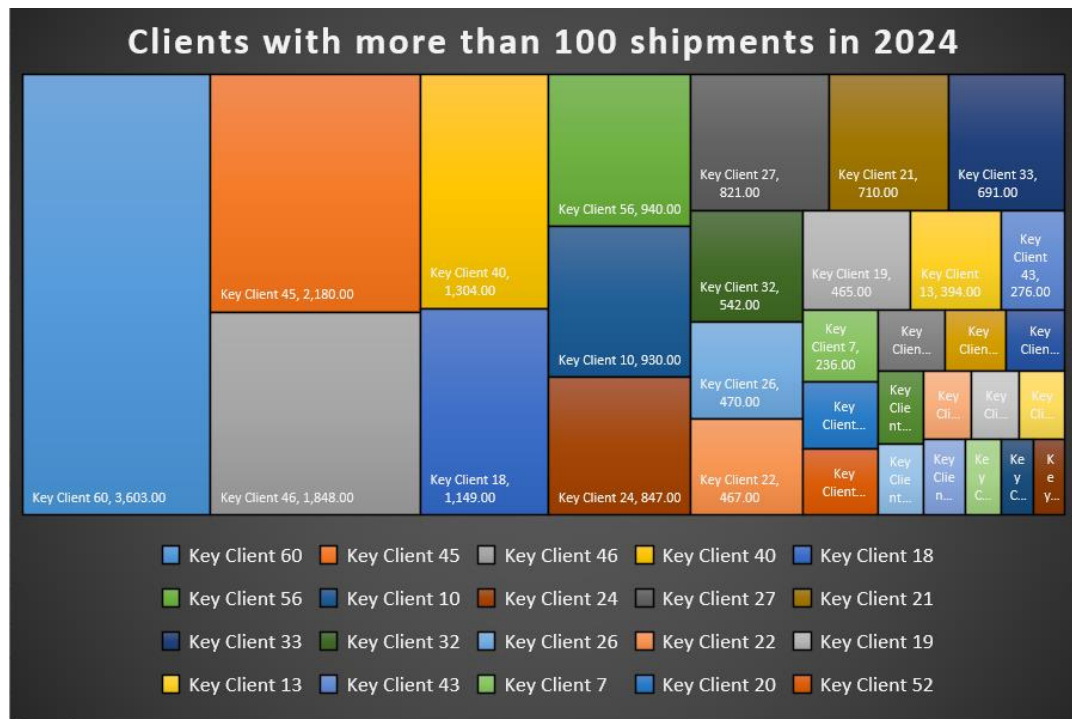
client_name,

YEAR(shipments.shipment_date) AS YearOfShipment,

```

COUNT(shipments.id) AS TotalShipments
FROM
    clients
LEFT JOIN
    shipments
ON
    clients.id = shipments.client_id
GROUP BY
    client_name,
    YEAR(shipments.shipment_date)
HAVING
    COUNT(shipments.id) > 100 AND
    YEAR(shipments.shipment_date) = 2024
ORDER BY
    TotalShipments DESC;

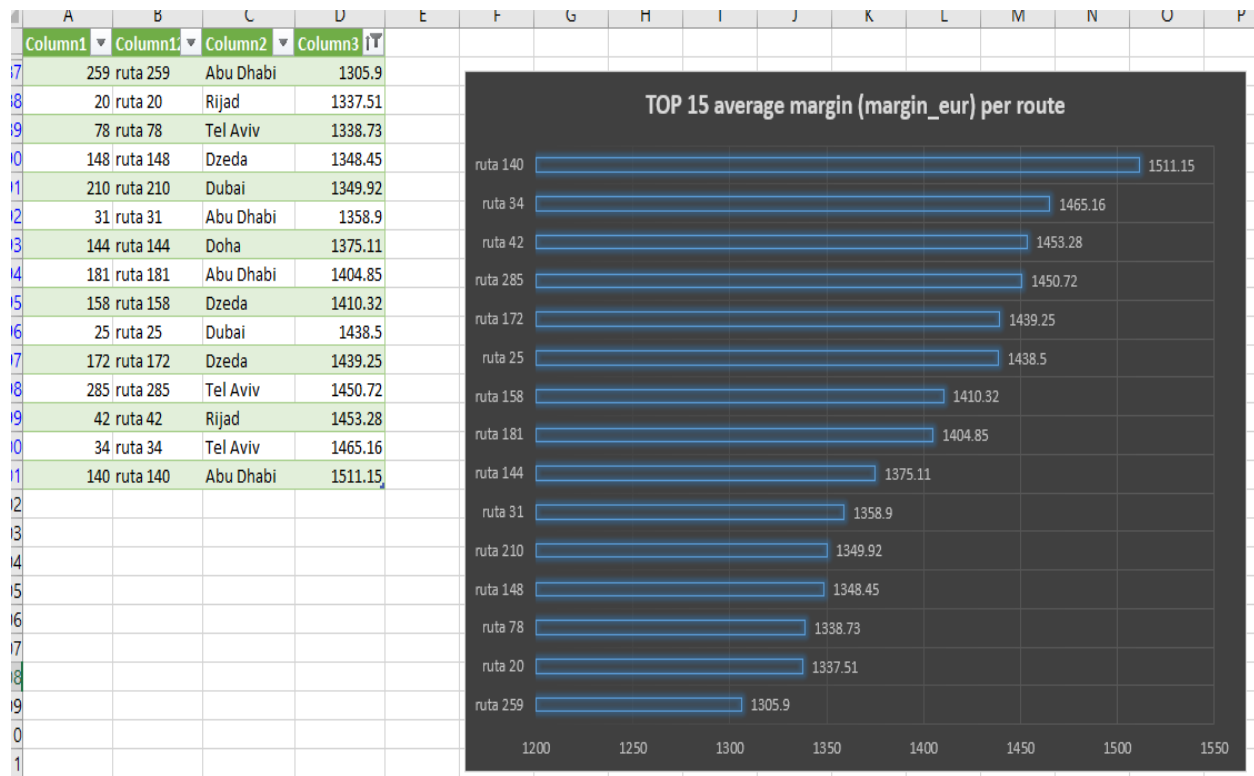
```



```
/* =====  
4. Koliki je prosečan margin_eur po ruti?  
===== */
```

```
-- Grupisati po route_id.  
-- Izračunati AVG(margin_eur).
```

```
SELECT  
    route_id,  
    city,  
    ROUND(AVG(margin_eur),2) AS Average_Margin  
FROM  
    locations  
LEFT JOIN  
    routes  
ON  
    locations.id = routes.destination_id  
LEFT JOIN  
    shipments  
ON  
    routes.id = shipments.route_id  
GROUP BY  
    route_id,  
    city  
ORDER BY  
    Average_Margin DESC;
```



/* =====

5. Koji tip vozila je najčešće korišten?

===== */

-- JOIN shipments i vehicles.

-- COUNT broj pošiljki po vehicle_type.

-- Sortirati opadajuće po broju pošiljki.

SELECT

vehicle_type,

COUNT(shipments.id) AS TotalShipments

FROM

vehicles

LEFT JOIN

shipments

ON

vehicles.id = shipments.vehicle_id

GROUP BY

vehicle_type

ORDER BY

TotalShipments DESC;

Vehicle Type	Total Shipments
Standard Truck	15770
Mega Trailer	10489
Reefer	10450
ADR	5291

/* =====

6. Kolika je ukupna margina po regionu unutar svake industrije?

===== */

-- JOIN shipments i routes i JOIN shipments i clients .

-- SUM(tons) po region PARTITION BY industry.

-- Grupisati po region.

SELECT

clients.industry,

routes.region,

ROUND(SUM(shipments.margin_eur),2) AS total_margin_eur

FROM

shipments

LEFT JOIN

routes

ON shipments.route_id = routes.id

LEFT JOIN

clients

ON shipments.client_id = clients.id

GROUP BY

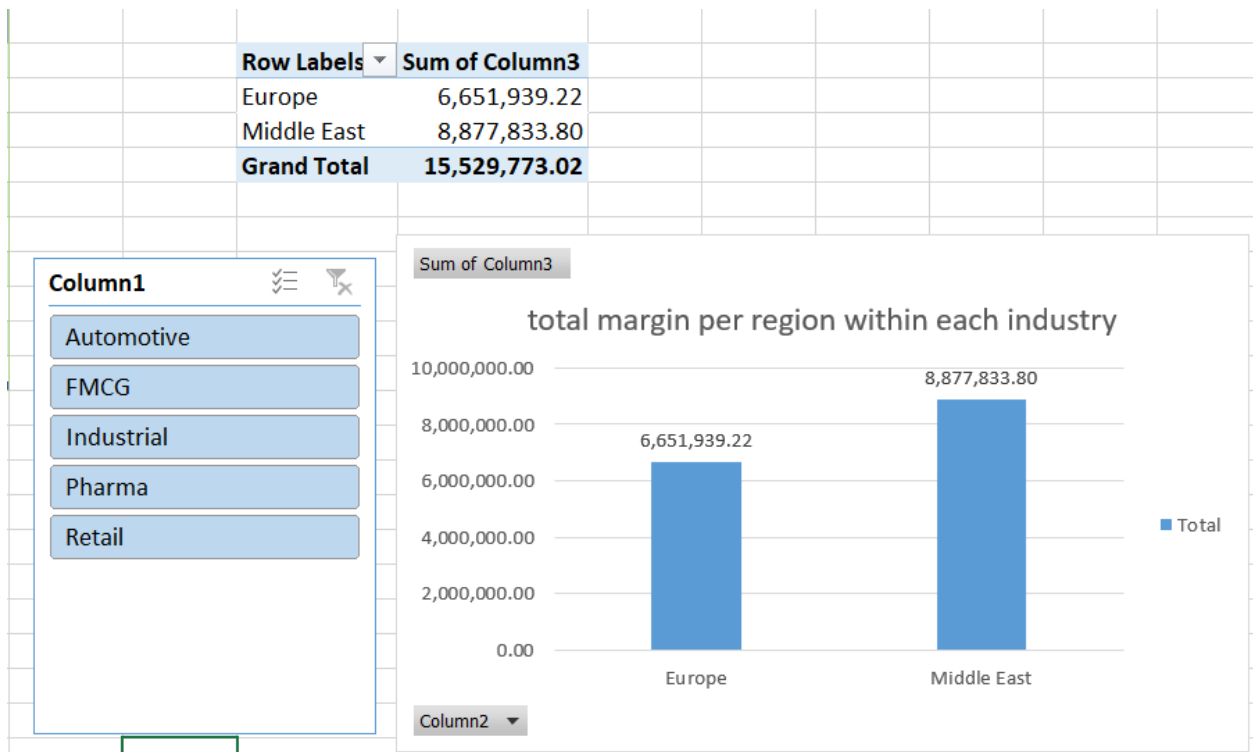
clients.industry,

routes.region

ORDER BY

clients.industry,

routes.region DESC;



/* =====

7. Koji su top 5 klijenata po prihodu po toni?

===== */

-- SUM(revenue_eur)/SUM(tons) po client_name.

-- Sortirati opadajuće.

-- Koristiti TOP 5.

SELECT TOP 5

client_name,

SUM(revenue_eur)/SUM(tons) AS RevenuePerTone

FROM

shipments

LEFT JOIN

clients

ON

shipments.client_id = clients.id

GROUP BY

client_name

ORDER BY

RevenuePerTone DESC;

1	top 5 clients by revenue per ton	REVENUE PER TONE
2	Key Client 34	776.6483024
3	Key Client 12	259.1574012
4	Key Client 44	252.045106
5	Key Client 53	241.1583094
6	Key Client 8	239.5347489
7		

/* =====

8. Koliki je prosječan revenue po pošiljci po godinama i mesecima?

===== */

-- Izračunati AVG(revenue_eur) iz shipments GROUP BY YEAR and MONTH.

SELECT

YEAR(shipments.shipment_date) AS Year,

MONTH(shipments.shipment_date) AS Month,

ROUND(AVG(revenue_eur),2) AS AverageRevenue

FROM

shipments

GROUP BY

YEAR(shipments.shipment_date),

MONTH(shipments.shipment_date)

ORDER BY

Year ASC,

Month ASC;



/* =====

9. Koji KAM upravlja najvećim brojem klijenata i koliku marginu po klijentu ima koji?

===== */

-- JOIN clients i key_account_managers.

-- COUNT broj klijenata po KAM-u.

-- Sortirati opadajuće po margini po klijentu.

WITH kam_margin AS (

SELECT

c.kam_id,

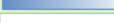
ROUND(SUM(s.margin_eur),2) AS total_margin

FROM clients c

LEFT JOIN shipments s

```
        ON c.id = s.client_id
    GROUP BY
        c.kam_id
),
kam_clients AS (
    SELECT
        kam_id,
        COUNT(*) AS number_of_clients
    FROM clients
    GROUP BY
        kam_id
)
```

```
SELECT
    k.name,
    kc.number_of_clients,
    km.total_margin,
    ROUND(km.total_margin * 1.0 / kc.number_of_clients, 2) AS margin_per_client
FROM key_account_managers k
LEFT JOIN kam_clients kc
    ON k.id = kc.kam_id
LEFT JOIN kam_margin km
    ON k.id = km.kam_id
ORDER BY
    margin_per_client DESC;
```

	A	B	C	D
1	Manager ▾	Numb_of_clients ▾	Total Margin ▾	MarginPerClient ▾
2	Sandra Kovac Andric	 4	4,078,855.62	1,019,713.91
3	Ivana Stojanovic	 7	2,003,438.87	286,205.55
4	Ana Petrovic	 13	3,189,018.48	245,309.11
5	Milica Nikolic	 9	2,054,520.36	228,280.04
6	Teodora Markovic	 11	2,383,087.17	216,644.29
7	Jelena Ilic	 10	1,195,808.25	119,580.82
8	Marija Jovanovic	 6	625,044.27	104,174.04
9				

/* =====

10. Koja relacija (origin → destination) ima najveću ukupnu kilometražu?

===== */

-- Koristiti tabelu routes.

-- SUM(distance_km) po origin_id i destination_id.

-- Sortirati opadajuće.

SELECT

r.id,

o.city AS origin_name,

d.city AS destination_name,

SUM(r.distance_km) AS Total_KM

FROM routes r

LEFT JOIN locations o

ON r.origin_id = o.id

LEFT JOIN locations d

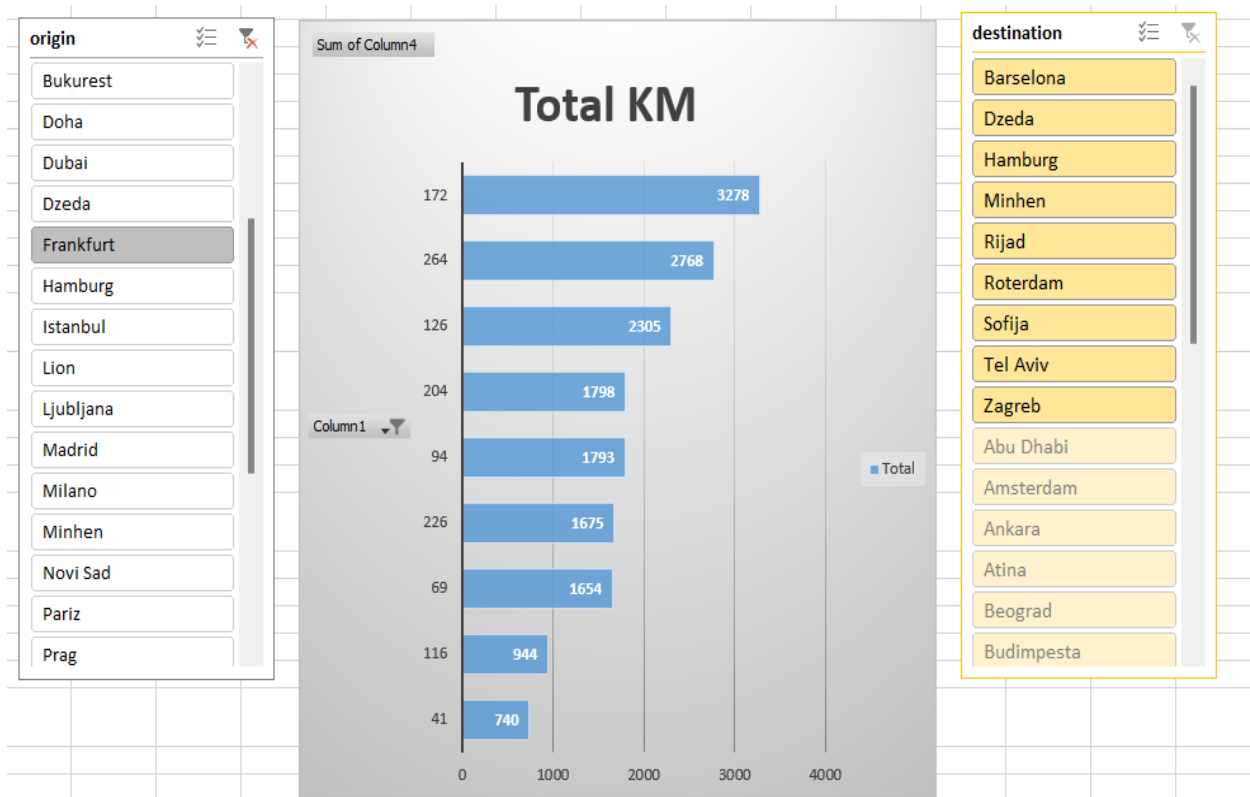
ON r.destination_id = d.id

GROUP BY

r.id,
o.city,
d.city

ORDER BY

Total_KM DESC;



```
/* =====
```

```
11. Ranking klijenata po prihodu unutar industrije
```

```
(Window Function)
```

```
===== */
```

```
-- Izračunati SUM(revenue_eur) po klijentu.
```

```
-- Koristiti RANK() ili DENSE_RANK().
```

```
-- PARTITION BY industry.
```

```
-- ORDER BY ukupni prihod DESC.
```

```
WITH client_revenue AS (
```

```
    SELECT
```

```
        c.id,
```

```
        c.client_name,
```

```
        c.industry,
```

```
        SUM(s.revenue_eur) AS TotalRevenue
```

```
    FROM
```

```
        clients c
```

```
    LEFT JOIN
```

```
        shipments s
```

```
    ON
```

```
        c.id = s.client_id
```

```
    GROUP BY
```

```
        c.id,
```

```
        c.client_name,
```

```
        c.industry
```

SELECT

cr.industry,

cr.client_name,

cr.TotalRevenue,

RANK() OVER(PARTITION BY cr.industry ORDER BY cr.TotalRevenue DESC) AS

Revenue_Rank

FROM

```
client_revenue cr
```

ORDER BY

industry,

Revenue_Rank;

Sum of revenue	Column Labels					
Row Labels	Automotive	FMCG	Industrial	Pharma	Retail	Grand Total
Key Client 10			4,698,859.94			4,698,859.94
Key Client 27					3,714,062.42	3,714,062.42
Key Client 46		8,817,583.93				8,817,583.93
Key Client 56				5,444,614.80		5,444,614.80
Key Client 60	15,680,172.56					15,680,172.56
Grand Total	15,680,172.56	8,817,583.93	4,698,859.94	5,444,614.80	3,714,062.42	38,355,293.66

```
/* =====  
12. Mesečni prihod i kumulativni prihod (Running Total)  
===== */
```

```
-- Grupisati po YEAR(shipment_date) i MONTH(shipment_date).  
-- Izračunati mjesečni revenue.  
-- Koristiti SUM() OVER (ORDER BY godina, mjesec)  
-- za kumulativni prihod.
```

```
WITH monthly_revenue AS (  
    SELECT  
        YEAR(shipment_date) AS Year,  
        MONTH(shipment_date) AS Month,  
        ROUND(SUM(revenue_eur), 2) AS TotalRevenue  
    FROM shipments  
    GROUP BY  
        YEAR(shipment_date),  
        MONTH(shipment_date)  
)
```

```
SELECT  
    Year,  
    Month,  
    TotalRevenue,  
    ROUND(  
        SUM(TotalRevenue) OVER (ORDER BY Year, Month), 2
```

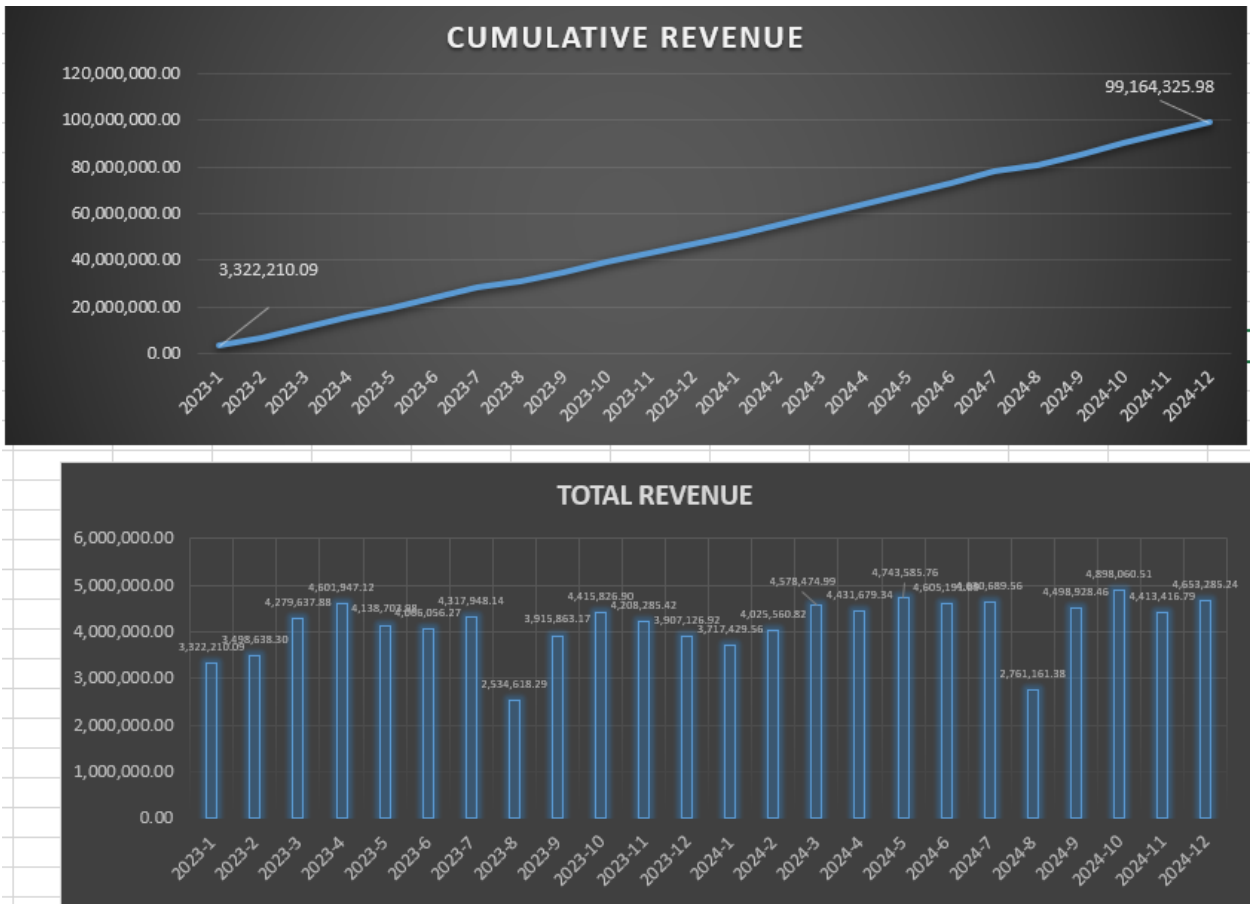
) AS CumulativeRevenue

FROM monthly_revenue

ORDER BY

Year,

Month;



/* =====

13. Year-over-Year (YoY) i Month-over-Month (MoM) rast prihoda

===== */

-- Izračunati godišnji prihod.

-- Koristiti LAG() da dohvatite prihod prethodne godine.

-- Izračunati procentualni rast.

WITH monthly_revenue AS (

SELECT

YEAR(shipment_date) AS Year,

MONTH(shipment_date) AS Month,

SUM(revenue_eur) AS TotalRevenue

FROM shipments

GROUP BY

YEAR(shipment_date),

MONTH(shipment_date)

)

SELECT

Year,

Month,

ROUND(TotalRevenue,2) AS TotalRevenue_TY,

ROUND(LAG(TotalRevenue, 12) OVER (ORDER BY Year, Month),2) AS TotalRevenue_LY,

ROUND(

(TotalRevenue - LAG(TotalRevenue, 12) OVER (ORDER BY Year, Month)) * 1.0

/ LAG(TotalRevenue, 12) OVER (ORDER BY Year, Month),

2

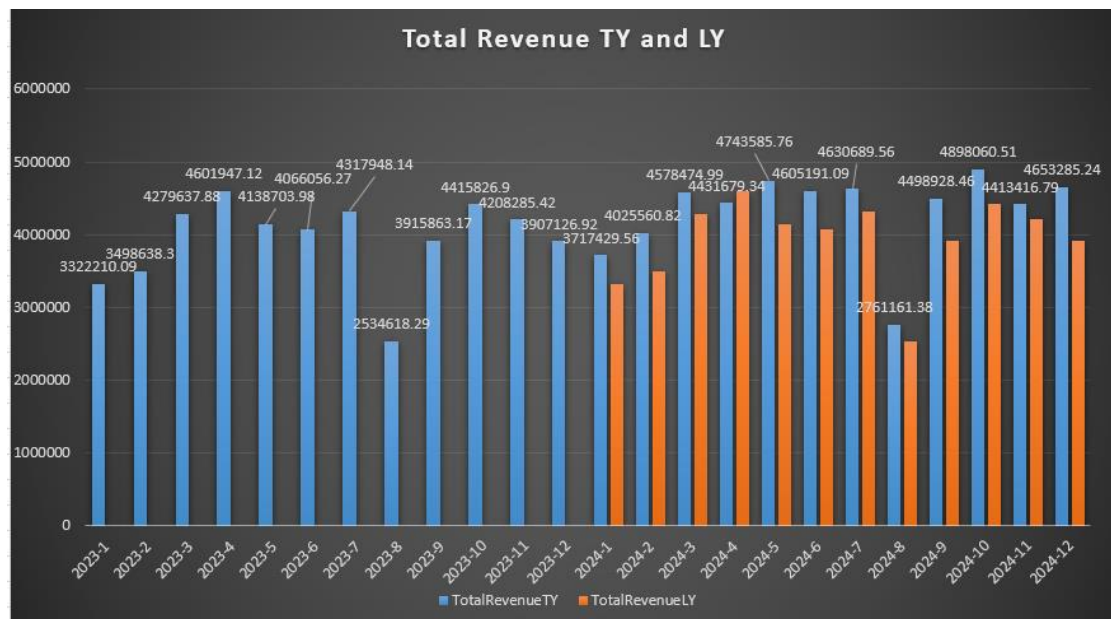
) AS YoY_Revenue_Growth,

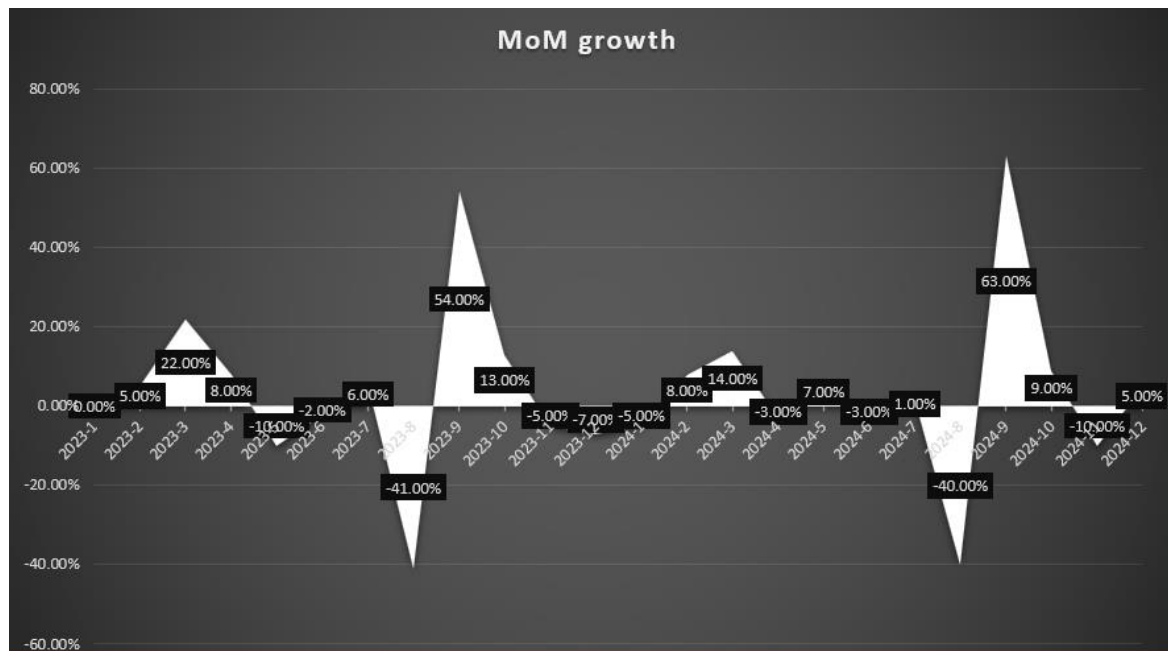
ROUND(LAG(TotalRevenue) OVER (ORDER BY Year, Month),2) AS
TotalRevenue_LastMonth,

```

ROUND(
    (TotalRevenue - LAG(TotalRevenue) OVER (ORDER BY Year, Month)) * 1.0
    / LAG(TotalRevenue) OVER (ORDER BY Year, Month),
    2
) AS MoM_Revenue_Growth
FROM monthly_revenue
ORDER BY
    Year,
    Month;

```





/* =====

14. Profit margin % po klijentu i odstupanje od prosjeka i njihov revenue_share % (učešće svakog klijenta u ukupnom prihodu)

===== */

-- Izračunati margin_pct = margin_eur / revenue_eur.

-- Izračunati prosječni margin_pct koristeći AVG() OVER ().

-- Prikazati razliku između klijenta i globalnog prosjeka.

WITH client_metrics AS (

SELECT

c.client_name,

SUM(s.revenue_eur) AS total_revenue,

SUM(s.margin_eur) AS total_margin

FROM clients c

```
LEFT JOIN shipments s
  ON c.id = s.client_id
GROUP BY c.client_name
),
calculated AS (
  SELECT
    client_name,
    total_revenue,
    total_margin,
    total_margin / NULLIF(total_revenue, 0) AS margin_pct
  FROM client_metrics
)
```

```
SELECT
  client_name,
  margin_pct,
  AVG(margin_pct) OVER () AS avg_margin_pct,
  margin_pct - AVG(margin_pct) OVER () AS deviation_from_avg,
  total_revenue / NULLIF(SUM(total_revenue) OVER (), 0) AS revenue_share
FROM calculated
ORDER BY deviation_from_avg DESC;
```

	A	B	C	D	E
1	Client	Margin pct	Avg margin pct	Difference from avg	revenue share of total
2	Key Client 4	19.17%	15.57%	3.60%	0.02%
3	Key Client 15	16.87%	15.57%	1.31%	0.05%
4	Key Client 50	16.78%	15.57%	1.22%	0.38%
5	Key Client 8	16.65%	15.57%	1.09%	0.37%
6	Key Client 37	16.27%	15.57%	0.70%	0.27%
7	Key Client 14	16.24%	15.57%	0.68%	0.17%
8	Key Client 58	16.24%	15.57%	0.68%	0.13%
9	Key Client 53	16.05%	15.57%	0.48%	0.88%
10	Key Client 59	15.99%	15.57%	0.42%	0.58%
11	Key Client 44	15.97%	15.57%	0.40%	0.53%
12	Key Client 41	15.96%	15.57%	0.39%	0.38%
13	Key Client 7	15.94%	15.57%	0.38%	1.09%
14	Key Client 24	15.89%	15.57%	0.32%	3.73%
15	Key Client 16	15.87%	15.57%	0.30%	0.34%
16	Key Client 26	15.85%	15.57%	0.28%	2.36%
17	Key Client 33	15.84%	15.57%	0.28%	3.09%
18	Key Client 43	15.83%	15.57%	0.27%	1.39%
19	Key Client 39	15.83%	15.57%	0.26%	0.39%
20	Key Client 38	15.80%	15.57%	0.23%	0.12%
21	Key Client 10	15.76%	15.57%	0.19%	4.74%
22	Key Client 27	15.75%	15.57%	0.18%	3.75%
23	Key Client 48	15.75%	15.57%	0.18%	0.72%
24	Key Client 52	15.73%	15.57%	0.17%	1.13%
25	Key Client 42	15.73%	15.57%	0.16%	0.66%
26	Key Client 19	15.70%	15.57%	0.13%	2.08%
27	Key Client 22	15.69%	15.57%	0.13%	2.17%
28	Key Client 21	15.69%	15.57%	0.12%	3.52%
29	Key Client 32	15.68%	15.57%	0.11%	2.47%
30	Key Client 46	15.67%	15.57%	0.11%	8.89%
31	Key Client 57	15.67%	15.57%	0.10%	0.24%
32	Key Client 12	15.66%	15.57%	0.09%	0.67%
33	Key Client 45	15.65%	15.57%	0.08%	9.51%
34	Key Client 60	15.65%	15.57%	0.08%	15.81%
35	Key Client 56	15.63%	15.57%	0.06%	5.49%
36	Key Client 28	15.58%	15.57%	0.02%	0.28%
37	Key Client 2	15.58%	15.57%	0.01%	0.85%
38	Key Client 18	15.55%	15.57%	-0.02%	5.61%
39	Key Client 31	15.53%	15.57%	-0.04%	0.58%
40	Key Client 40	15.50%	15.57%	-0.07%	5.70%
41	Key Client 36	15.47%	15.57%	-0.10%	0.54%
42	Key Client 3	15.40%	15.57%	-0.17%	0.54%
43	Key Client 23	15.40%	15.57%	-0.17%	0.19%
44	Key Client 49	15.37%	15.57%	-0.20%	0.35%
45	Key Client 13	15.35%	15.57%	-0.22%	1.89%
46	Key Client 25	15.30%	15.57%	-0.27%	0.76%
47	Key Client 47	15.28%	15.57%	-0.29%	0.24%
48	Key Client 9	15.27%	15.57%	-0.29%	0.77%
49	Key Client 1	15.21%	15.57%	-0.36%	0.62%
50	Key Client 35	15.15%	15.57%	-0.41%	0.86%
51	Key Client 20	15.14%	15.57%	-0.43%	0.89%
52	Key Client 17	15.10%	15.57%	-0.46%	0.15%
53	Key Client 11	15.06%	15.57%	-0.50%	0.21%
54	Key Client 30	15.01%	15.57%	-0.55%	0.27%
55	Key Client 54	15.00%	15.57%	-0.57%	0.03%
56	Key Client 55	14.71%	15.57%	-0.86%	0.33%
57	Key Client 29	14.61%	15.57%	-0.96%	0.08%
58	Key Client 5	14.48%	15.57%	-1.09%	0.08%
59	Key Client 6	14.16%	15.57%	-1.41%	0.02%
60	Key Client 51	13.01%	15.57%	-2.55%	0.04%
61	Key Client 34	12.86%	15.57%	-3.21%	0.01%

```
/* =====  
15. Najprofitabilnija ruta po regionu (Top 1 po regionu)  
===== */
```

```
-- Koristiti CTE za agregaciju margin_eur po ruti i regionu.
```

```
-- Koristiti ROW_NUMBER().
```

```
-- PARTITION BY region.
```

```
-- ORDER BY ukupna margina DESC.
```

```
-- Filtrirati gdje je row_number = 1.
```

```
WITH region_margin_per_route AS (  
    SELECT  
        region,  
        route_id,  
        SUM(margin_eur) AS TotalMargin  
    FROM  
        routes  
    LEFT JOIN  
        shipments  
    ON  
        routes.id = shipments.route_id  
    GROUP BY
```

```

        region,
        route_id
    ),
    ranked_routes AS (
    SELECT
        region,
        route_id,
        TotalMargin,
        ROW_NUMBER() OVER (PARTITION BY region ORDER BY TotalMargin DESC) AS rn
    FROM
        region_margin_per_route
    )
SELECT
    region,
    route_id,
    TotalMargin
FROM ranked_routes
WHERE rn = 1;

```

	A	B	C
1	Region ▼	Rute ▼	Margin ▼
2	Europe	220	54,355.93
3	Middle East	285	219,058.17
4			